A GROUPOID OVER A FIELD HAS THE DIMENSION OF ITS STABILIZER

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span. The equidimensionality input from the uncarded group-side dimension lemma is declared as external.

Allowed standard prerequisites (editorial).

- Equidimensionality of R and G. lemma-groupoid-on-field-locally-finite-type-dimension (same chapter, carded L26644) and lemma-group-over-field-locally-finite-type-dimension (same chapter, NOT carded, declared external). Reduces the claim to dim_e(R) = dim_e(G).
- Unique base field in a variety ring. Varieties, proposition unique-base-field (cross-chapter, named). Identifies the integral closures k_a and k_b of the two field images.

Source reference. Original lemma 06FE: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1160–1160. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04LU: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1107–1107. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01T6: focused-seeds/geometry/S9-smg-groupoid-on-field-dimension-equal-stabilizer/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01OZ: focused-seeds/geometry/S9-smg-groupoid-on-field-dimension-equal-stabilizer/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06DY: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 859–859. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Source reference. Original reference 03XG: focused-seeds/geometry/S9-smg-groupoid-on-field-dimension-equal-stabilizer/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0468: focused-seeds/geometry/S9-smg-groupoid-on-field-dimension-equal-stabilizer/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0464: focused-seeds/geometry/S9-smg-groupoid-on-field-dimension-equal-stabilizer/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

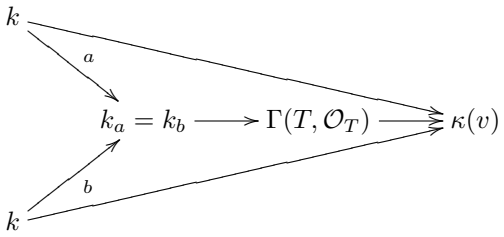
Source reference. Original reference 04MF: focused-seeds/geometry/S9-smg-groupoid-on-field-dimension-equal-stabilizer/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04MK: focused-seeds/geometry/S9-smg-groupoid-on-field-dimension-equal-stabilizer/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 06DY assume s, t are locally of finite type. Let $G = \mathrm{Spec}(k) \times_{\Delta, \mathrm{Spec}(k) \times_B \mathrm{Spec}(k), t \times s} R$ be the stabilizer group algebraic space. Then we have $\dim(R) = \dim(G)$.*

Proof. Since G and R are equidimensional (see Lemmas 04LU and 06FE) it suffices to prove that $\dim_e(R) = \dim_e(G)$. Let V be an affine scheme, $v \in V$, and let $\varphi : V \to R$ be an étale morphism of schemes such that $\varphi(v) = e$. Note that V is a Noetherian scheme as $s \circ \varphi$ is locally of finite type as a composition of morphisms locally of finite type and as V is quasi-compact (use Morphisms of Spaces, Lemmas 03XG, 0468, and 0464 and Morphisms, Lemma 01T6). Hence V is locally connected (see Properties, Lemma 01OZ and Topology, Lemma 04MF). Thus we may replace V by the connected component containing v (it is still affine as it is an open and closed subscheme of V). Set $T = V_{red}$ equal to the reduction of V. Consider the two morphisms $a, b : T \to \mathrm{Spec}(k)$ given by $a = s \circ \varphi|_T$ and $b = t \circ \varphi|_T$. Note that a, b induce the same field map $k \to \kappa(v)$ because $\varphi(v) = e$! Let $k_a \subset \Gamma(T, \mathcal{O}_T)$ be the integral closure of $a^\sharp(k) \subset \Gamma(T, \mathcal{O}_T)$. Similarly, let $k_b \subset \Gamma(T, \mathcal{O}_T)$ be the integral closure of $b^\sharp(k) \subset \Gamma(T, \mathcal{O}_T)$. By Varieties, Proposition 04MK we see that

$k_a = k_b$. Thus we obtain the following commutative diagram



As discussed above the long arrows are equal. Since $k_a = k_b \to \kappa(v)$ is injective we conclude that the two morphisms a and b agree. Hence $T \to R$ factors through G. It follows that $R_{red} = G_{red}$ in an open neighbourhood of e which certainly implies that $\dim_e(R) = \dim_e(G)$. □